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Life has trickled back to a 'lonely, dead river'

Author: Hume, Mark

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Abstract: "Acid mines last a long time," Mr. [Jack Minard] said. "The Romans mined on the Thames River and that site is still generating acid rock drainage, 3,000 years later."

Full text: mhume@globeandmail.com To restore the river they first had to go to the top of the mountain. It was a long, exhausting journey that involved countless planning meetings, but in the end the Tsolum River Restoration Society succeeded in winning provincial government support for a $4.5-million project to seal an old mine on Mount Washington that was generating acid rock drainage. Now the river is springing back to life. The Mt. Washington Copper Mining Co. only operated from 1964 through 1966, excavating copper ore on a small 13 hectare site, near Courtenay, on Vancouver Island. But when the bankrupt company abandoned the mine, it left behind 940,000 tonnes of waste rock - and a toxic legacy that for the next 44 years brewed sulphuric acid, which in turn leached copper and other heavy metals from the slag. The pollution went into Pyrrhotite and Murex Creeks, and from there into the Tsolum River. Salmon runs were already depressed by the mid-1960s because of logging damage. The acid rock drainage kept them from recovering, by impairing the development of young salmon and by forcing adult salmon to turn away from the river when they returned to spawn. "The fish would come into the river but then they would back out . . .they would avoid the system," says Jack Minard, executive director of the Tsolum River Restoration Society. By 1985, the once productive salmon river had become vacant of life. Instead of 200,000 pink salmon, 11,000 chum and 15,000 Coho, the river was down to a handful of fish. Some years no salmon came at all. And a run of 3,500 steelhead, which had reached sizes of 10 kilograms, became so rare they were thought extinct. "It was this lonely, dead river," Mr. Minard said. Dead but not forgotten. Starting in the 1980s, local volunteers, led initially by Father Charles Brandt, began to work at restoring the river with the help of the federal Department of Fisheries and Oceans. At first it was thought hatcheries could restock the system. But the runs never recovered, no matter how many young salmon were introduced. One year 2.5 million pink salmon were released in the river. None came back. In 1985, a study identified the problem of acid rock drainage. The river looked beautiful, but the water was toxic. In an effort to stop the old mine from leaching poison into the river, the waste rock was gathered into a gigantic pile, mixed with lime to neutralize the acid, and covered with gravel and clay. Water quality improved, and salmon did better, but copper concentrations remained too high for fish to flourish. Searching for solutions, the Tsolum River Partnership (a coalition that involved provincial and federal government agencies, TimberWest, the Mining Association of B.C., and the Tsolum River Restoration Society) discussed possibly flooding the mine site to seal the acid-generating ore underwater; capping the mine with concrete; digging trenches to funnel water away from the site; or collecting the runoff and sending it through a pipeline to a treatment plant. But most options offered only partial solutions or weren't practical. The treatment plant, for example, would have needed to operate for centuries. "Acid mines last a long time," Mr. Minard said. "The Romans mined on the Thames River and that site is still generating acid rock drainage, 3,000 years later." In 2008, the provincial government, inspired in part by a personal donation of $50,000 from Bob Hager, approved a plan to cap the Mt. Washington site with a waterproof seal, made of polyester impregnated with bitumen. The material is rolled out like a giant carpet. It locks out water and oxygen, which generates sulphuric acid when it contacts sulphide bearing ore. The seal went on in 2009-2010 with immediate and stunning results. "We have not exceeded our target of 11 parts per billion for dissolved copper in the river since the membrane was installed," Mr. Minard said. "We have seen a substantial change. The best way to say it is now water quality is not an issue." For more than 40 years the Tsolum was toxic. Now clouds of aquatic insects, long vanished from the river, can be seen dancing over the water while schools of salmon fry dart below. Last year 1,000 adult Coho returned to spawn. "That is amazing. We haven't had those numbers since 1959," Mr. Minard said. Steelhead, thought to be extinct, are in the river again. Chum and pink salmon runs haven't rebounded yet, but young fish are thriving. Much work to repair logging damage still needs to be done, and a way to provide adequate flows in summer has to be found, but Mr. Minard is ecstatic. "It's working," he said. "The river is coming back."

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